

Tutorial 10

Hierarchical Regression and Correlation

Statistical Computation and Analysis
Spring 2026

Tutorial Outline

- Hierarchical regression
- Regression versus correlation
- Pearson's r , population correlation ρ , and ρ^2
- Covariance matrix
- Simple Bayesian bivariate Normal correlation model

Main idea

This tutorial focuses on two ways statistical models represent dependence in data.

Grouped dependence

Observations from the same group may be more similar.

For example:
houses nested within counties

Modeling response:
hierarchical regression

Symmetric association

Two variables may vary together without assigning one as outcome and the other as predictor.

For example:
uranium and log radon

Modeling response:
correlation model

Dataset

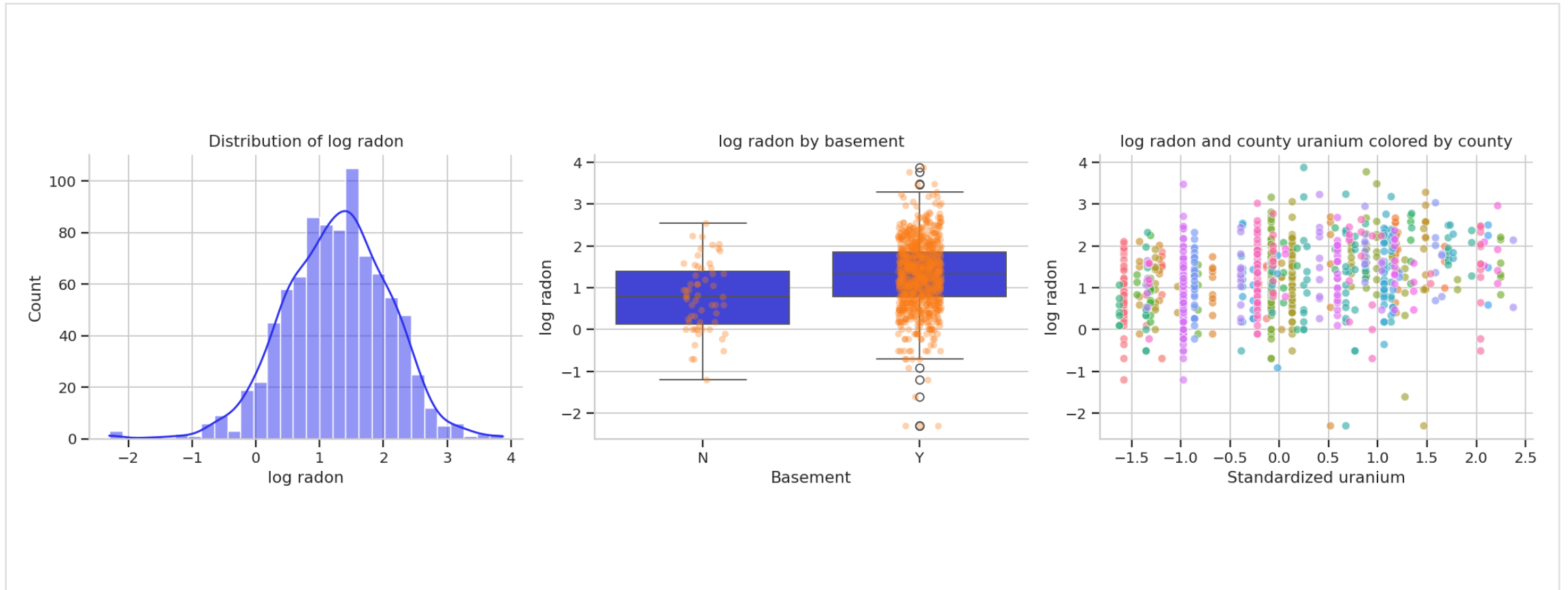
PART 1: HIERARCHICAL REGRESSION

- Dataset: Household-level radon measurements from Minnesota.
- Dependent variable: log_radon.
- Predictors: basement status and county-level uranium.
- Grouping variable: county.
- After cleaning: 878 observations from 85 counties.

Variable	Role
log_radon	Outcome
basement	House-level predictor
<i>uranium^z</i>	Standardized uranium predictor
county	Group-level structure

Note: uranium is standardized so the slope is interpreted per one standard deviation increase.

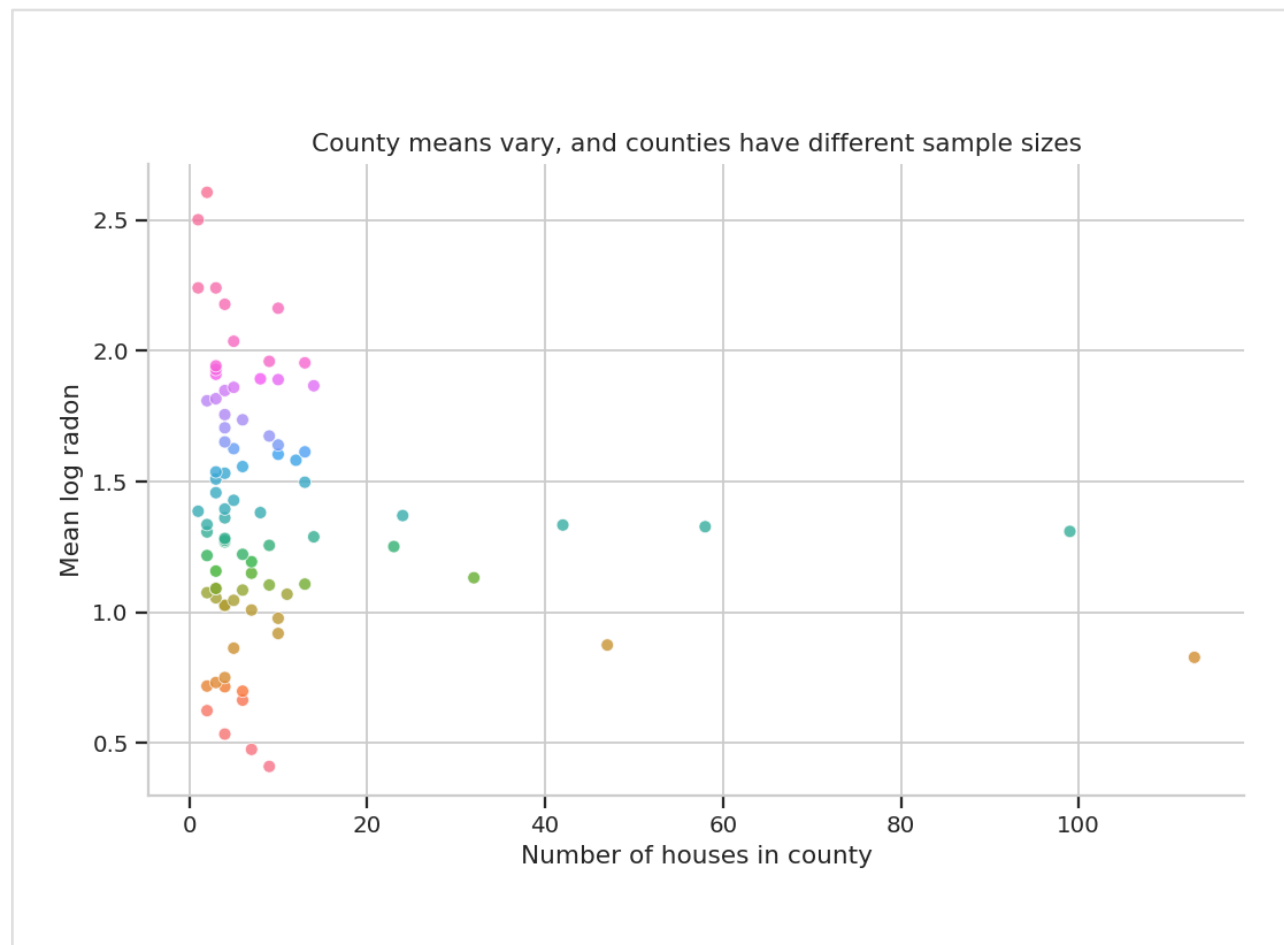
Inspect the data



Output from the notebook: distribution, basement comparison, and uranium association.

Use these plots to check the outcome scale, the basement groups, and the relation between uranium and log radon before fitting models.

County structure



County means vary and counties have different sample sizes.

- Houses are nested within counties.
- County means vary.
- Some counties have very few observations.
- This motivates a county-level intercept in the model.

Model 1 - Multiple regression

$$\log_radon \sim basement + uranium^z$$

$$y_i \sim Normal(\mu_i, \sigma)$$

$$\mu_i = \beta_0 + \beta_{basement} x_{basement,i} + \beta_{uranium^z} x_{uranium^z,i}$$

- β_0 : expected log radon for the reference basement category at average uranium.
- $\beta_{basement}$: expected difference between basement categories.
- $\beta_{uranium}$: expected change for one-SD increase in uranium.
- σ : residual house-level variation.

Model 1 - Multiple regression

$$\log_radon \sim basement + uranium^z$$

$$y_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \beta_0 + \beta_{basement} x_{basement,i} + \beta_{uranium^z} x_{uranium^z,i}$$

- *Modeling with Bambi:*

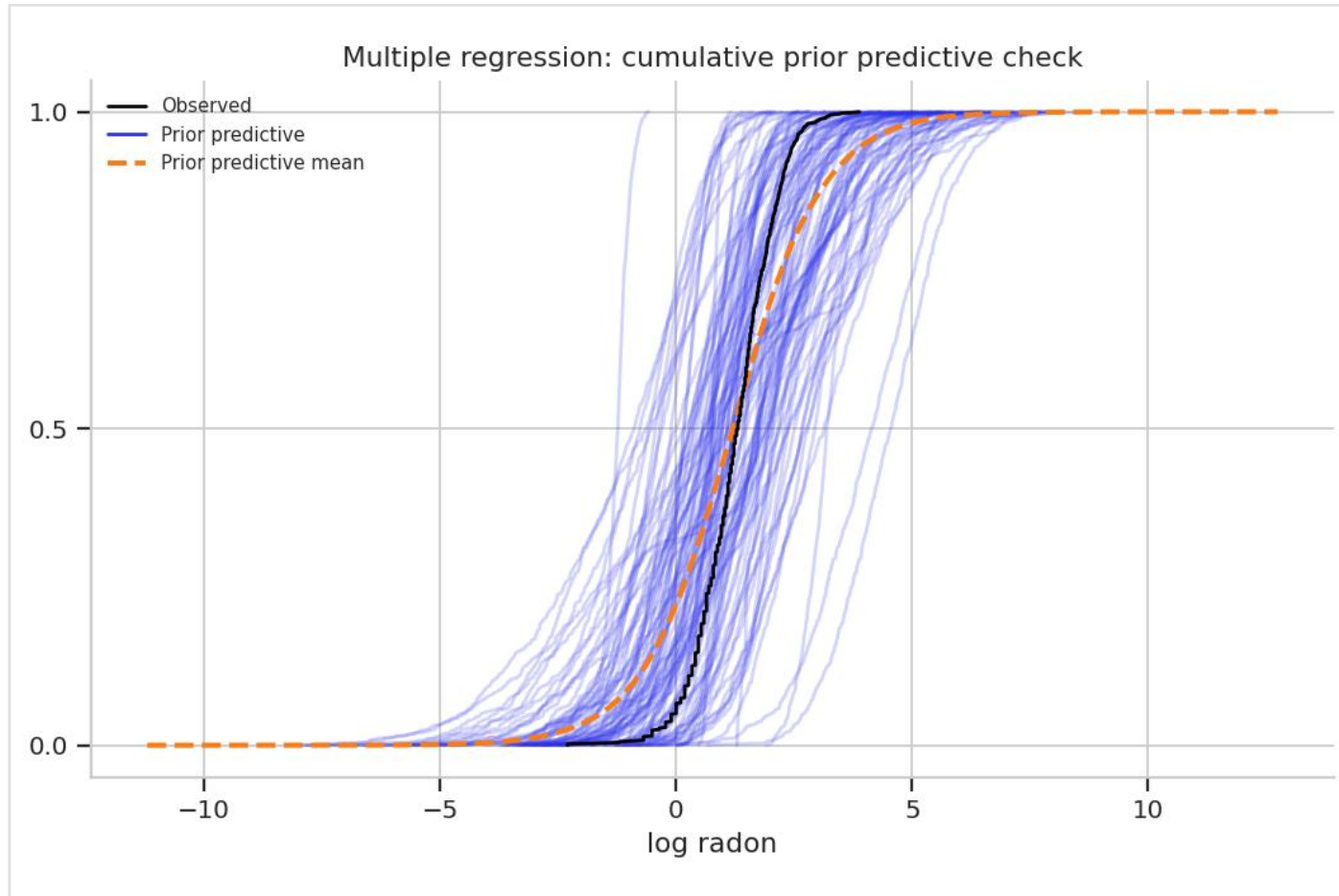
```
multiple_model = bmb.Model(  
    "log_radon ~ basement + uranium_z",  
    data=radon,  
    family="gaussian",  
    priors=radon_regression_priors,  
)  
print(multiple_model)
```

```
Formula: log_radon ~ basement + uranium_z  
Family: gaussian  
Link: mu = identity  
Observations: 878  
Priors:  
target = mu  
Common-level effects  
Intercept ~ Normal(mu: 1.2, sigma: 1.0)  
basement ~ Normal(mu: 0.0, sigma: 1.0)  
uranium_z ~ Normal(mu: 0.0, sigma: 1.0)  
  
Auxiliary parameters  
sigma ~ HalfNormal(sigma: 1.0)
```

- *Choosing our own priors (optional):*

```
radon_regression_priors = {  
    "Intercept": bmb.Prior("Normal", mu=1.2, sigma=1.0),  
    "basement": bmb.Prior("Normal", mu=0, sigma=1.0),  
    "uranium_z": bmb.Prior("Normal", mu=0, sigma=1.0),  
    "sigma": bmb.Prior("HalfNormal", sigma=1.0),  
}
```

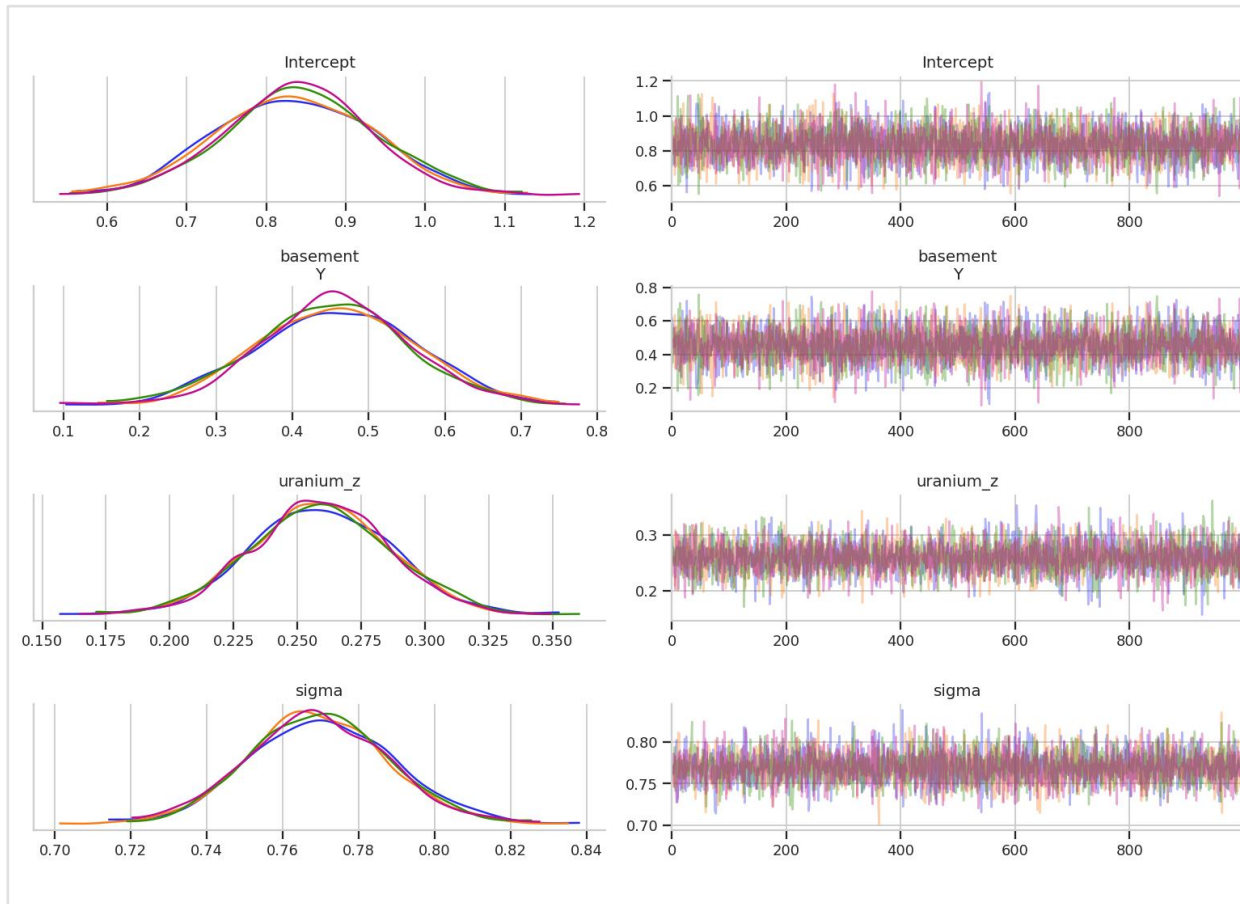
Multiple regression: prior predictive check



Cumulative prior predictive check.

- Checks whether the priors generate plausible log-radon values before fitting.
- Not a model-fit check.
- The observed distribution should fall within the range implied by the prior predictive distribution.

Multiple regression: sampling diagnostics



Trace plots for fixed effects and residual σ .

Parameter	ESS bulk	R-hat
Intercept	4301	1.00
basement[Y]	4319	1.00
<i>uranium^z</i>	4215	1.00
sigma	4348	1.00

- Diagnostics check sampling validity.
- Here the chains mixed well and R-hat is approximately 1.

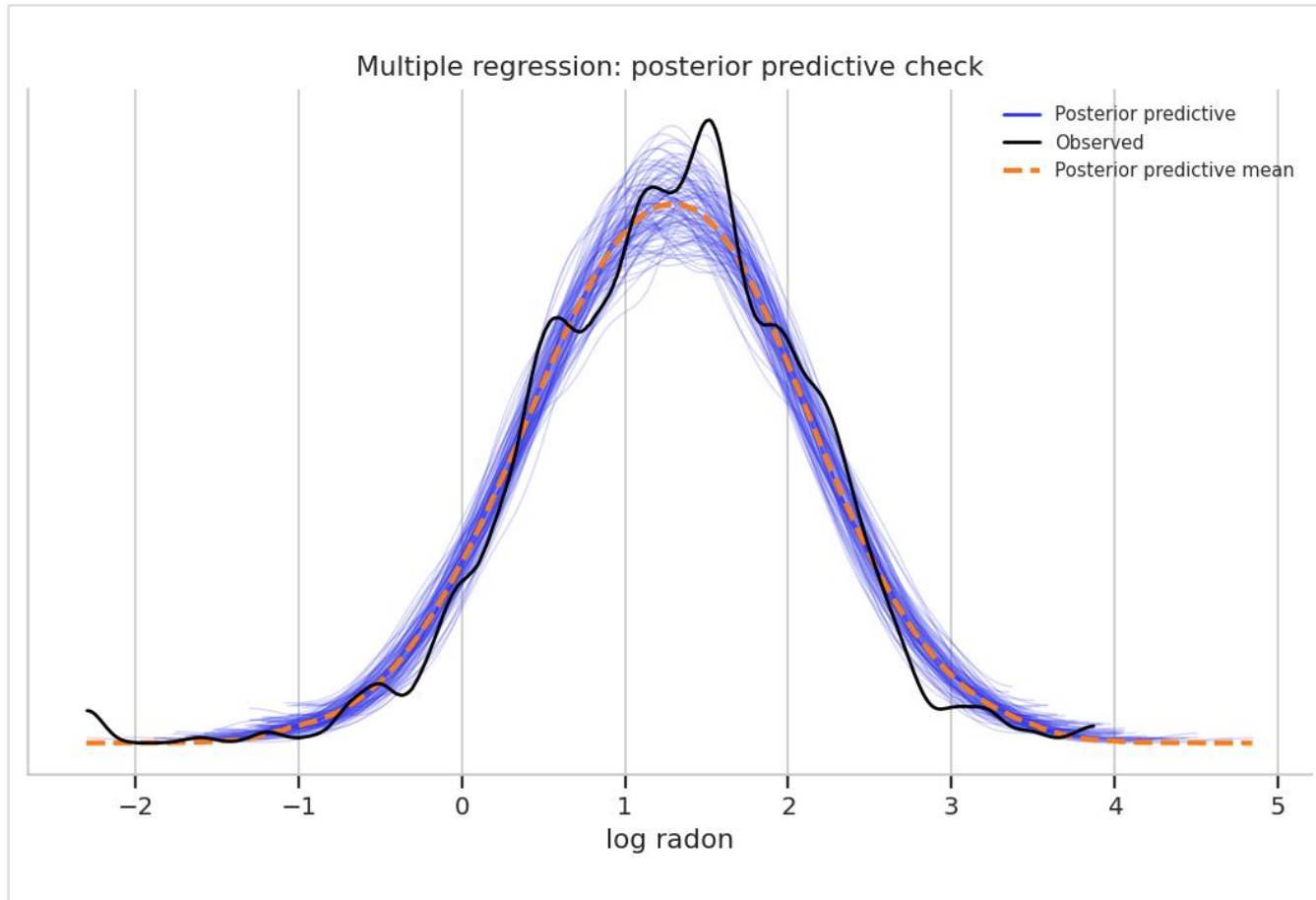
Multiple regression: posterior summary

Parameter	Mean	94% HDI
<i>Intercept</i>	0.838	[0.662, 1.025]
<i>basement</i> [Y]	0.456	[0.259, 0.640]
<i>uranium</i> ^z	0.259	[0.207, 0.309]
<i>sigma</i>	0.770	[0.735, 0.805]

- Houses with basement = Y have higher expected log radon than the reference group, after accounting for uranium.
- Higher county uranium is associated with higher expected log radon.
- σ is the remaining residual variation around the regression mean.

Interpretation should remain observational. The regression model describes association under the model assumptions.

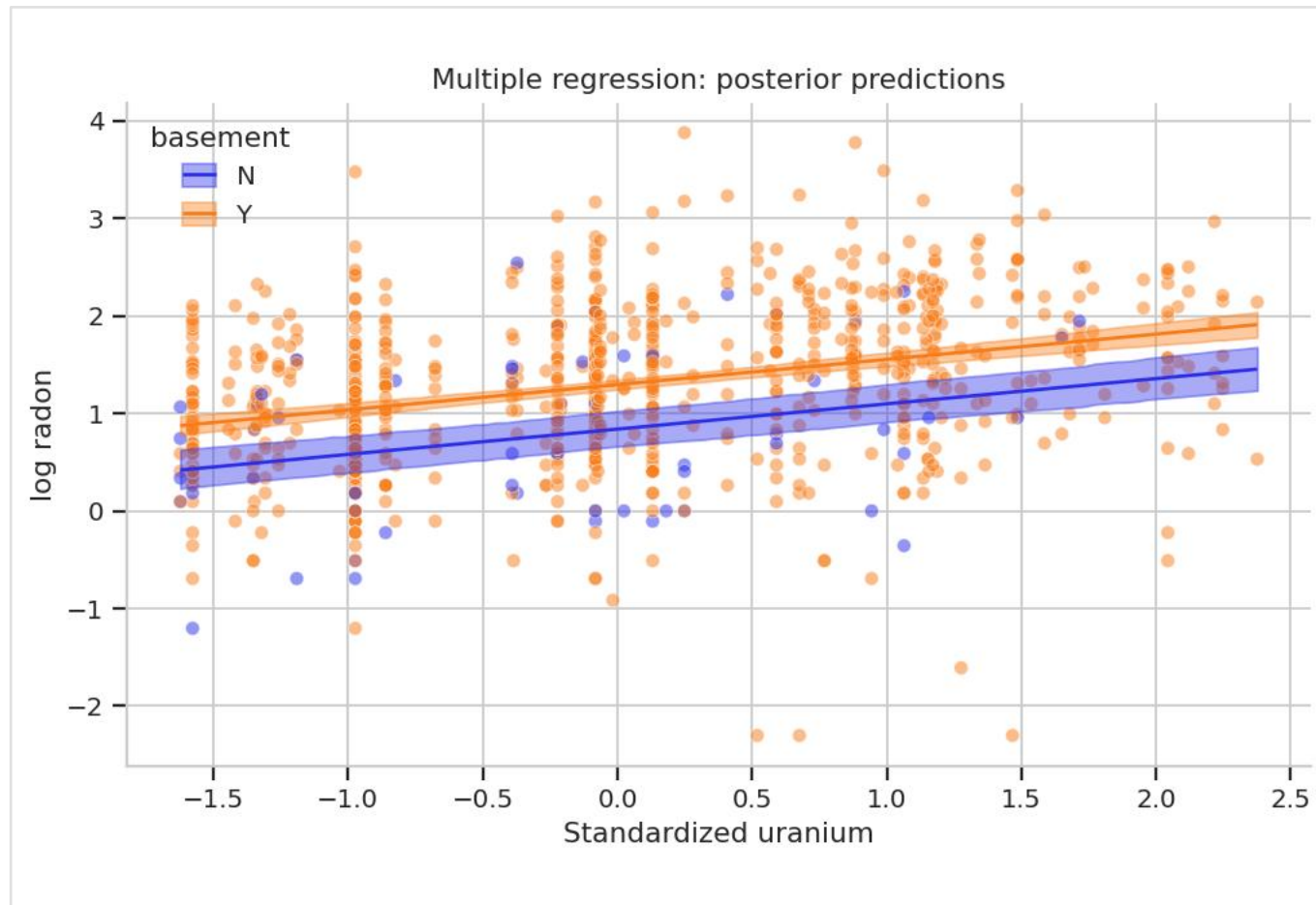
Multiple regression: posterior predictive check



Posterior predictive check for log radon.

- Posterior predictive checks ask whether the fitted model can generate data that resemble the observed data.
- The model captures the central part of the distribution reasonably well.
- Any mismatch in tails or shape suggests model limitations.

Multiple regression: posterior predictions



Predicted log radon as a function of standardized uranium and basement status.

- The fitted lines summarize expected log radon across uranium values.
- Basement status changes the intercept.
- The model uses one uranium slope shared across basement groups.

Why add a hierarchical county intercept?

The multiple regression treats all remaining variation as residual house-level noise.

$$\mu_i = \beta_0 + \beta_{\text{basement}} x_{\text{basement},i} + \beta_{\text{uranium}^z} x_{\text{uranium}^z,i}$$

But homes from the same county may share geological or environmental factors not fully captured by uranium.

Add a county-level baseline:

$$\mu_i = \beta_0 + \beta_{\text{basement}} x_{\text{basement},i} + \beta_{\text{uranium}^z} x_{\text{uranium}^z,i} + u_{\text{county}[i]}$$

Where $u_{\text{county}[i]}$ is random intercept adjustment for a specific county

Why add a hierarchical county intercept?

The multiple regression treats all remaining variation as residual house-level noise.

$$\mu_i = \beta_0 + \beta_{\text{basement}} x_{\text{basement},i} + \beta_{\text{uranium}^z} x_{\text{uranium}^z,i}$$

But homes from the same county may share geological or environmental factors not fully captured by uranium.

Add a county-level baseline:

$$\mu_i = \beta_0 + \beta_{\text{basement}} x_{\text{basement},i} + \beta_{\text{uranium}^z} x_{\text{uranium}^z,i} + u_{\text{county}[i]}$$

Where $u_{\text{county}[i]}$ is random intercept adjustment for a specific county

It is called random not because the county effect is “random noise” or meaningless, but because the model treats the county-specific intercepts as draws from a shared population distribution.

Model 2 - Hierarchical regression

$$\log_radon \sim basement + uranium^z + (1|county)$$

- $(1|county)$: each county gets an intercept adjustment.
- The county effects are modeled as related, not completely independent.
- σ_{county} measures between-county baseline variation.
- σ measures residual house-level variation.

$$y_i \sim Normal(\mu_i, \sigma)$$

$$\mu_i = (\beta_0 + u_{county[i]}) + \beta_{basement} x_{basement,i} + \beta_{uranium} x_{uranium,i}$$

$$u_j \sim Normal(0, \sigma_{county})$$

Model 2 - Hierarchical regression

$$\log_radon \sim basement + uranium^z + (1|county)$$

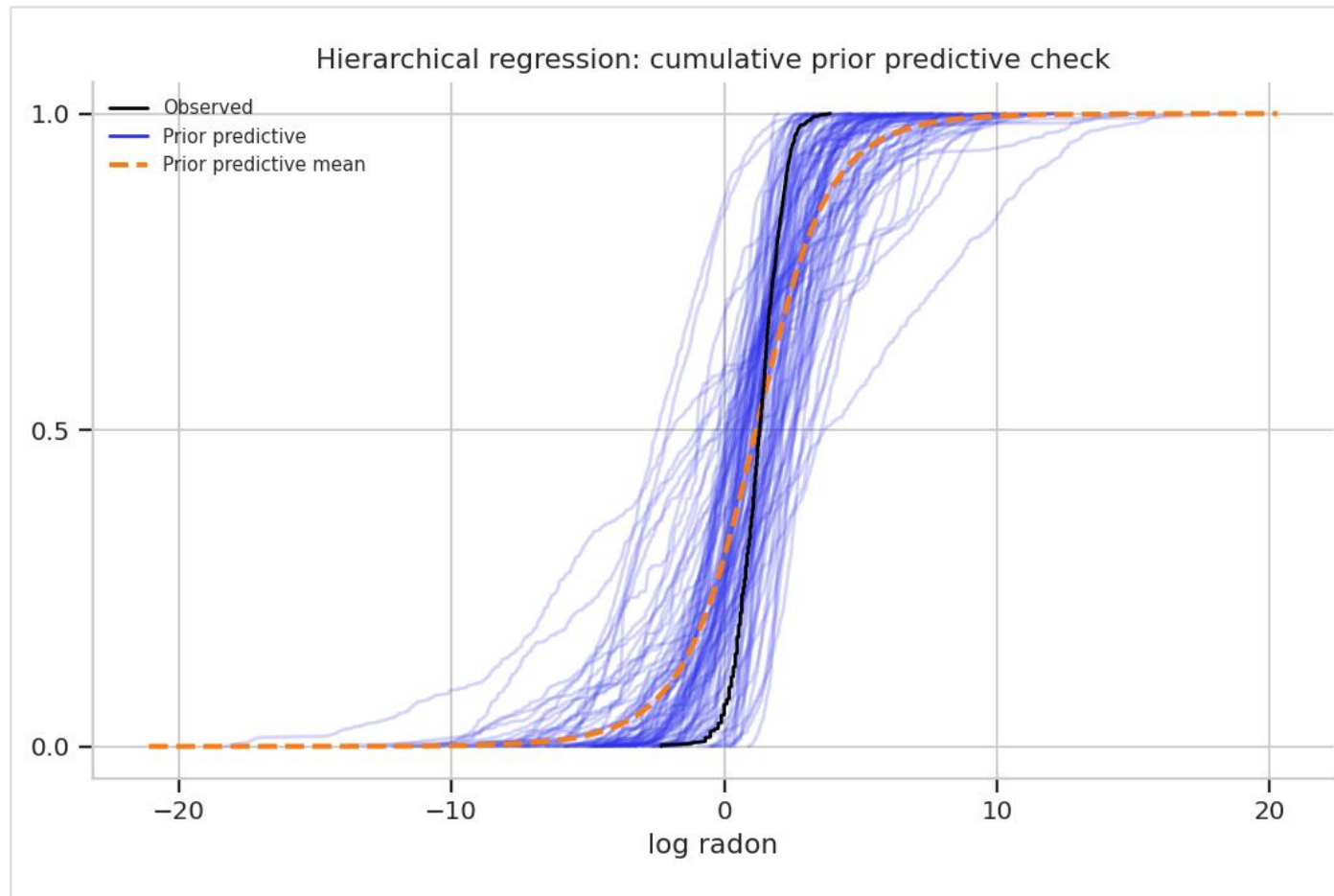
$$y_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = (\beta_0 + u_{county[i]}) + \beta_{basement} x_{basement,i} + \beta_{uranium} x_{uranium,i}$$

```
hierarchical_model = bmb.Model(  
    "log_radon ~ basement + uranium_z + (1|county)",  
    data=radon,  
    family="gaussian",  
    priors=radon_regression_priors,  
)  
print(hierarchical_model)
```

```
Formula: log_radon ~ basement + uranium_z + (1|county)  
Family: gaussian  
Link: mu = identity  
Observations: 878  
Priors:  
target = mu  
Common-level effects  
Intercept ~ Normal(mu: 1.2, sigma: 1.0)  
basement ~ Normal(mu: 0.0, sigma: 1.0)  
uranium_z ~ Normal(mu: 0.0, sigma: 1.0)  
  
Group-level effects  
1|county ~ Normal(mu: 0.0, sigma: HalfNormal(sigma: 2.0511))  
  
Auxiliary parameters  
sigma ~ HalfNormal(sigma: 1.0)
```

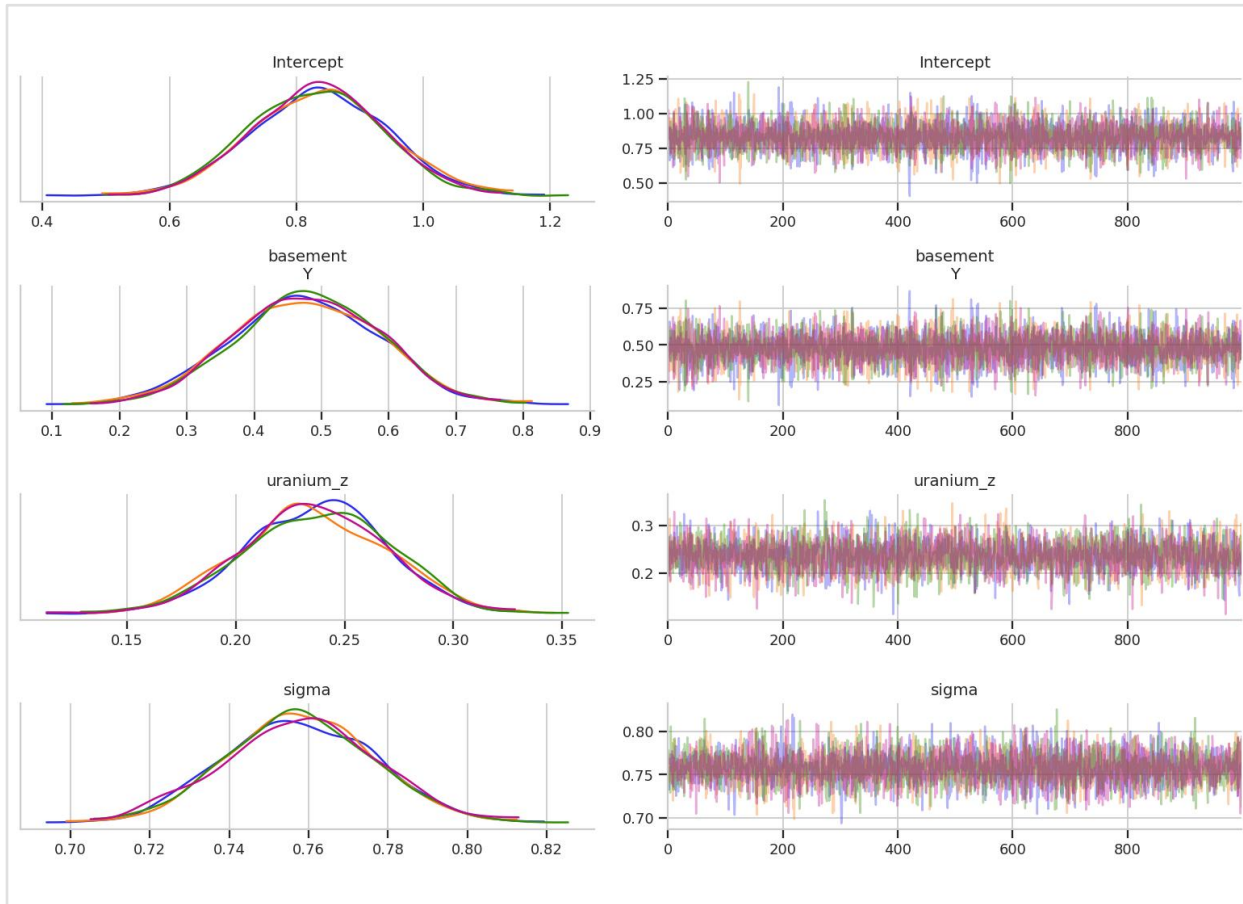
Hierarchical regression: prior predictive check



Cumulative prior predictive check.

The hierarchical model has an additional source of variation. The prior should allow county differences without generating impossible values too often. The check asks whether the model is coherent before seeing the data.

Hierarchical regression: diagnostics

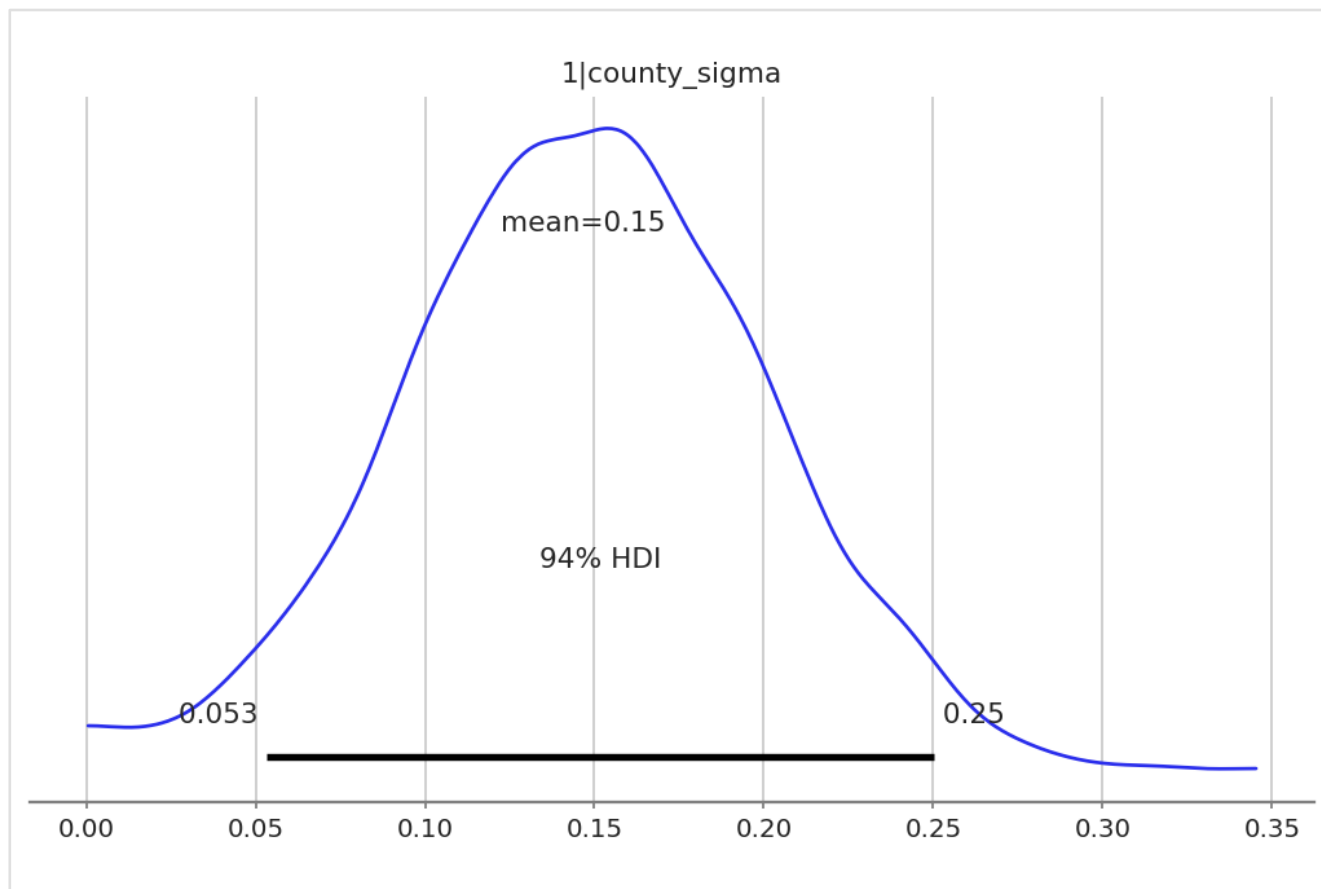


Trace plots for common parameters.

Parameter	ESS bulk	R-hat
Intercept	8327	1.00
<i>basement</i> [Y]	8141	1.00
<i>uranium</i> ^z	4504	1.00
sigma	5643	1.00

The common parameters show good sampling behavior. The county-level scale should also be checked before interpretation.

County-level variation

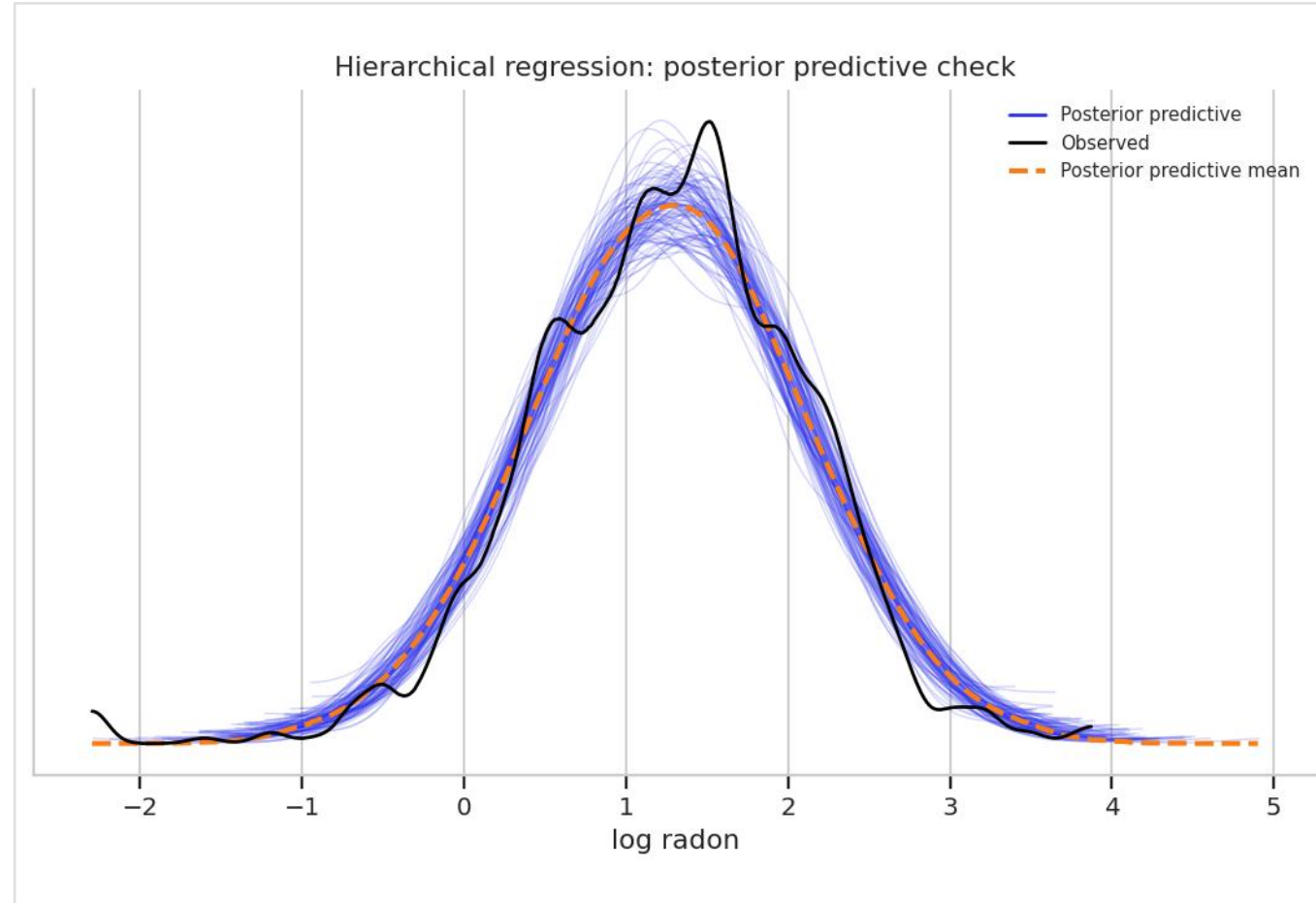


Posterior for the county-level standard deviation.

Parameter	Mean	94% HDI
$1 county_sigma$	0.147	[0.053, 0.250]

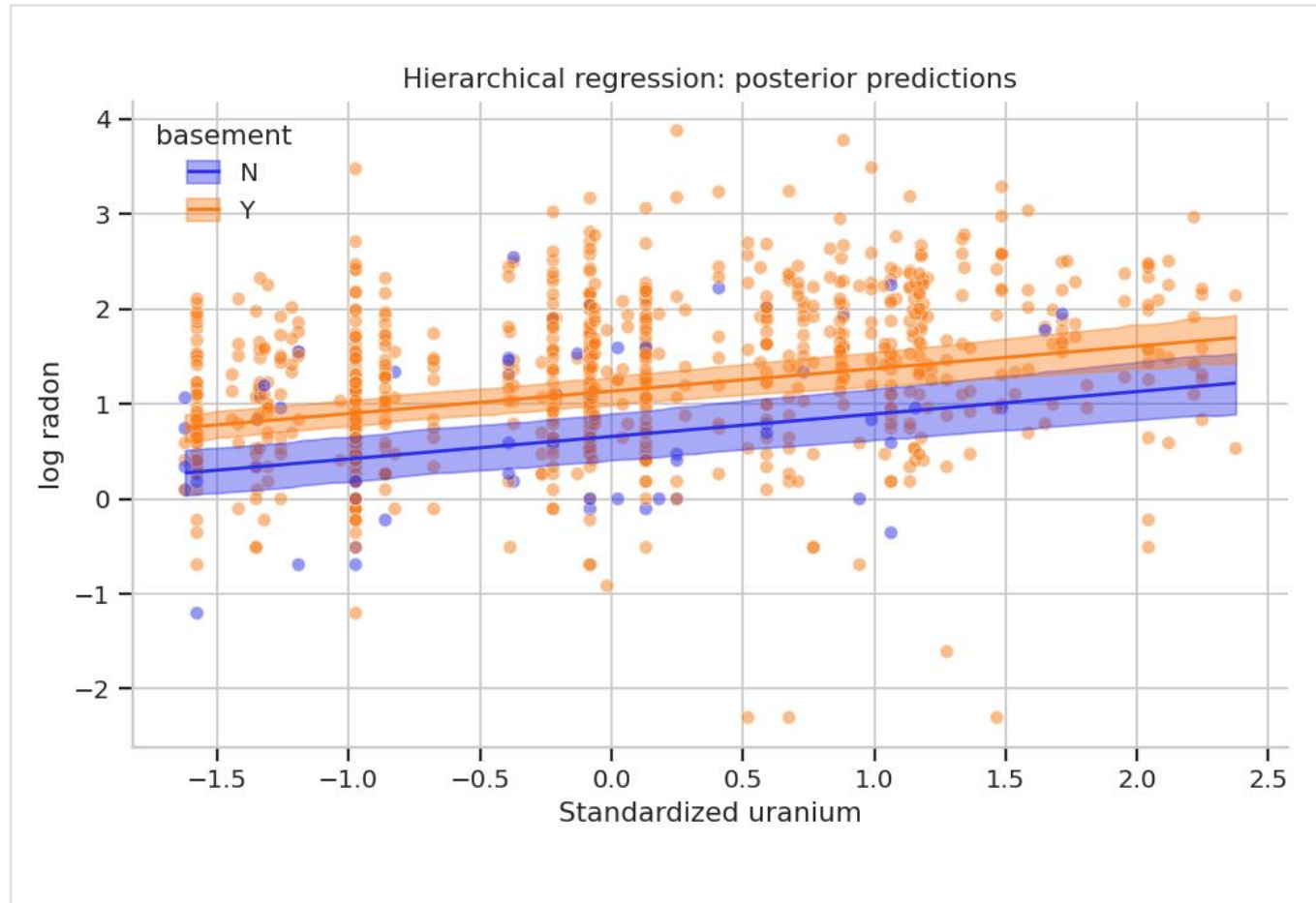
- This parameter measures variation in county-level intercepts.
- Values away from zero indicate remaining county-level baseline differences.
- It should be interpreted relative to the *residual* σ .

Hierarchical regression: posterior predictive check



Posterior predictive check for hierarchical regression.

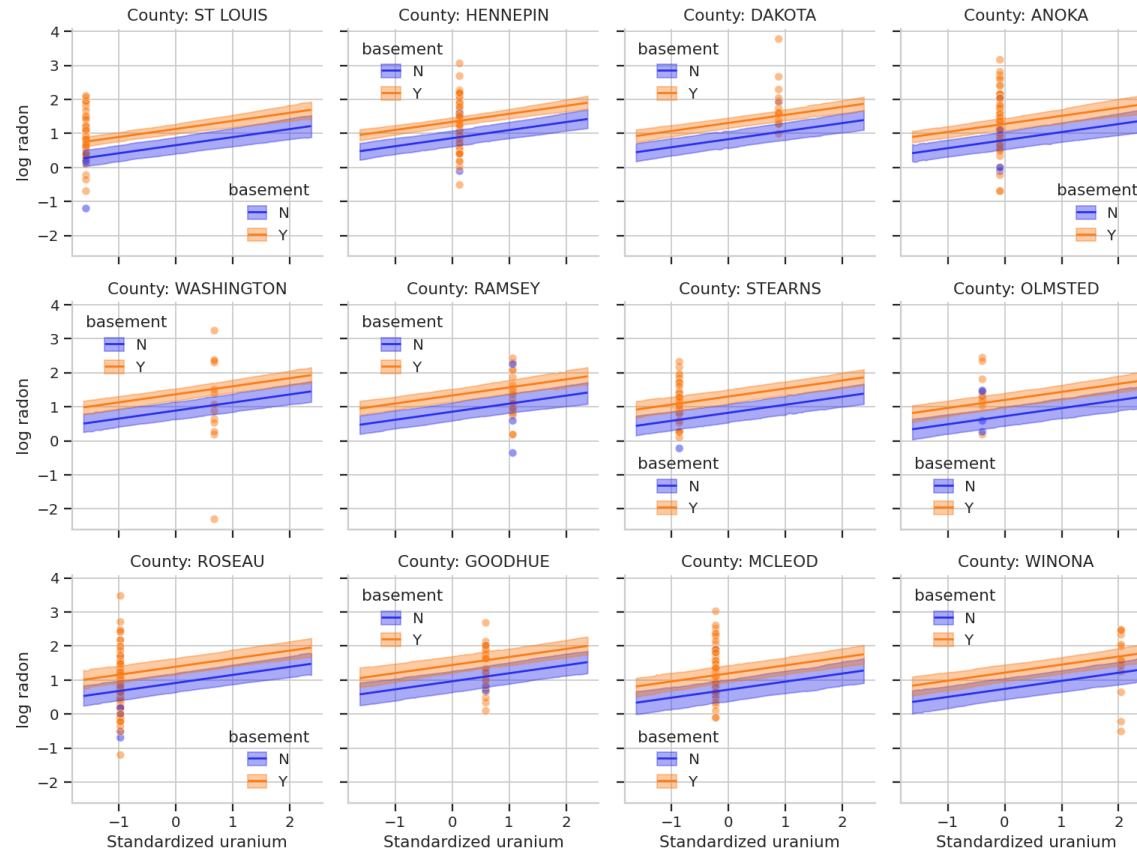
Hierarchical regression: posterior predictions



Predicted log radon as a function of uranium and basement status.

- The plotted lines summarize predictions averaged over the hierarchical structure.

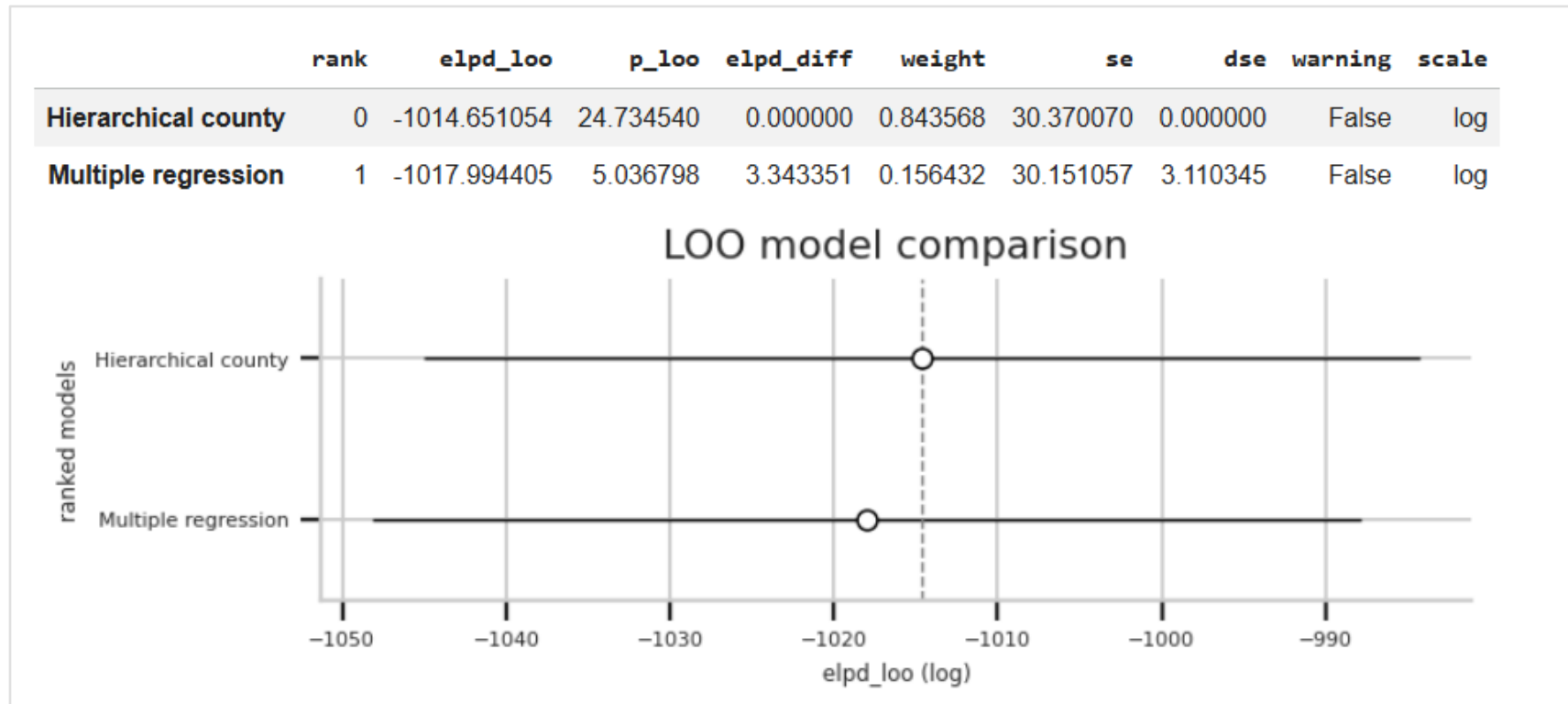
Hierarchical regression: posterior predictions



Predicted log radon as a function of uranium and basement status in each county.

- The plotted lines summarize predictions averaged over the hierarchical structure.
- The model also contains county-specific intercept deviations.
- The same predictors are interpreted after accounting for county baseline variation.

Short model comparison with LOO



LOO comparison plot.

- LOO estimates expected out-of-sample predictive accuracy.
- The hierarchical model is ranked first, but the difference is modest relative to uncertainty.
- The comparison asks about prediction, not whether the model is “true.”

Posterior estimates across models

Model	Parameter	Mean	94% HDI
Multiple	<i>basement</i> [Y]	0.456	[0.259, 0.640]
Multiple	<i>uranium</i> ^z	0.259	[0.207, 0.309]
Multiple	sigma	0.770	[0.735, 0.805]
Hierarchical	<i>basement</i> [Y]	0.477	[0.273, 0.674]
Hierarchical	<i>uranium</i> ^z	0.237	[0.176, 0.298]
Hierarchical	sigma	0.758	[0.722, 0.793]

- Adding county-level structure changes the estimates slightly.
- The uranium slope is somewhat smaller after accounting for county intercepts.
- The *residual* σ is also slightly smaller.
- Interpretation should combine diagnostics, PPC, and model comparison.

Regression section summary

- A multiple regression can include more than one predictor.
- A hierarchical regression can account for observations nested within groups.
- In the radon data, houses are nested within counties.
- The county random intercept separates county-level baseline variation from residual house-level variation.
- LOO asks whether the added structure improves expected out-of-sample prediction.

From regression to correlation

PART 2: CORRELATION AND COVARIANCE

The second part changes the scientific question.

Regression question

How does the expected value of y change as x changes?

One variable is the outcome.
The other variable is a predictor.

Model is directional.

Correlation question

How do two variables vary together?

Neither variable is privileged as the outcome.
Both are modeled jointly.

Model is symmetric.

Directional model versus correlational model

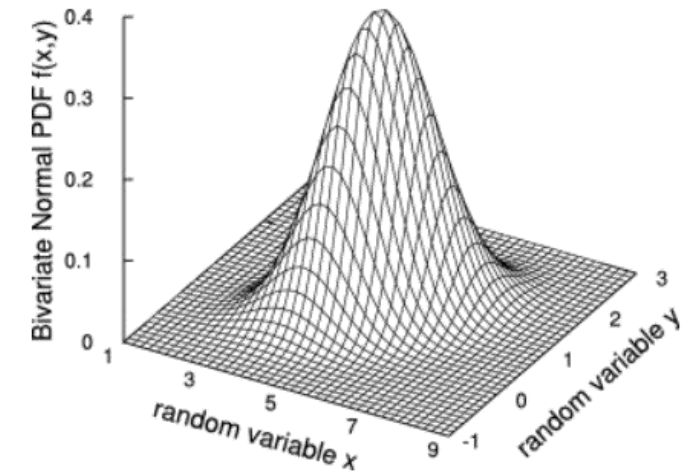
Regression model

$$y_i \sim \text{Normal}(\alpha + \beta x_i, \sigma)$$

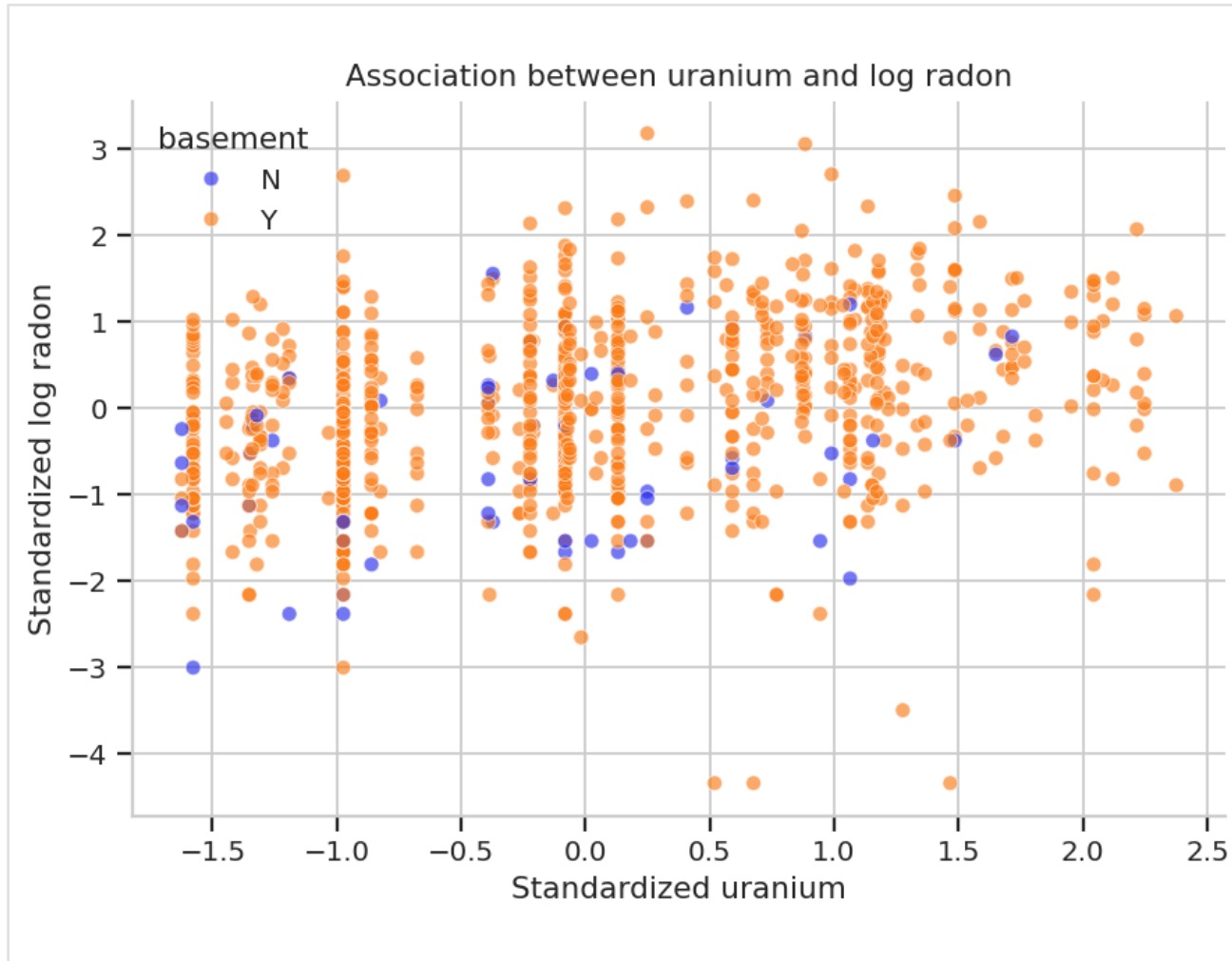
Correlation model

$$[x_i, y_i]^T \sim \text{MVNormal}(\mu, \Sigma)$$

- Regression imposes a predictor/outcome structure.
- Correlation models joint variation.
- Correlation alone is not a causal claim.



Inspect the association



Association between standardized uranium and standardized log radon.

- The scatterplot shows positive association.
- The plotted variables are standardized.
- Basement is shown only for context; the correlation model below uses two continuous variables.

Pearson correlation

$$r = \text{sample Pearson correlation}$$

- Measures the linear relationship between two datasets.
- r estimates the *population correlation* ρ .
- r is bounded between -1 and 1 .
- The frequentist p-value tests $H_0: \rho = 0$.

The correlation coefficient is calculated as follows:

$$r = \frac{\sum (x - m_x)(y - m_y)}{\sqrt{\sum (x - m_x)^2 \sum (y - m_y)^2}}$$

where m_x is the mean of the vector x and m_y is the mean of the vector y .

Under the assumption that x and y are drawn from independent normal distributions (so the population correlation coefficient is 0), the probability density function of the sample correlation coefficient r is ([1], [2]):

Pearson correlation

`scipy.stats.`

`pearsonr`

`pearsonr(x, y, *, alternative='two-sided', method=None, axis=0)` [\[source\]](#)

Pearson correlation coefficient and p-value for testing non-correlation.

The Pearson correlation coefficient [\[1\]](#) measures the linear relationship between two datasets. Like other correlation coefficients, this one varies between -1 and +1 with 0 implying no correlation. Correlations of -1 or +1 imply an exact linear relationship. Positive correlations imply that as x increases, so does y. Negative correlations imply that as x increases, y decreases.

This function also performs a test of the null hypothesis that the distributions underlying the samples are uncorrelated and normally distributed. (See Kowalski [\[3\]](#) for a discussion of the effects of non-normality of the input on the distribution of the correlation coefficient.) The p-value roughly indicates the probability of an uncorrelated system producing datasets that have a Pearson correlation at least as extreme as the one computed from these datasets.

Population correlation ρ

$$-1 \leq \rho \leq 1$$

- ρ is the population correlation parameter.
- In Bayesian inference, ρ has a prior and a posterior distribution.
- Positive values mean that high values of one variable tend to occur with high values of the other.
- Negative values mean the opposite.

Bayesian target:
 $p(\rho \mid data)$

- Instead of only reporting a point estimate, the Bayesian model gives uncertainty over plausible values of ρ .

ρ^2 as shared variance

$\rho^2 = \text{fraction of variance shared between two variables}$

- ρ^2 is a **normalized effect-size measure for correlation**.
- Example: if $\rho = 0.6$, then $\rho^2 = 0.36$.
- This means about 36% shared variance in the linear association sense.
- It is not a causal statement.

Pearson r and ρ^2

```
pearson_r, pearson_p = stats.pearsonr(radon["uranium_z"], radon["log_radon_z"])

pearson_table = pd.DataFrame({
    "quantity": ["Pearson r", "r squared", "p-value for H0: rho = 0"],
    "value": [pearson_r, pearson_r ** 2, pearson_p],
})

display(pearson_table)
```

	quantity	value
0	Pearson r	3.243355e-01
1	r squared	1.051935e-01
2	p-value for H0: rho = 0	5.926686e-23



The p-value is not the effect size.
The effect size is better summarized by r or r^2 .

Covariance and covariance matrix

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix}$$

$$\sigma_{xy} = \rho \sigma_x \sigma_y \rho = \frac{\sigma_{xy}}{(\sigma_x \sigma_y)}$$

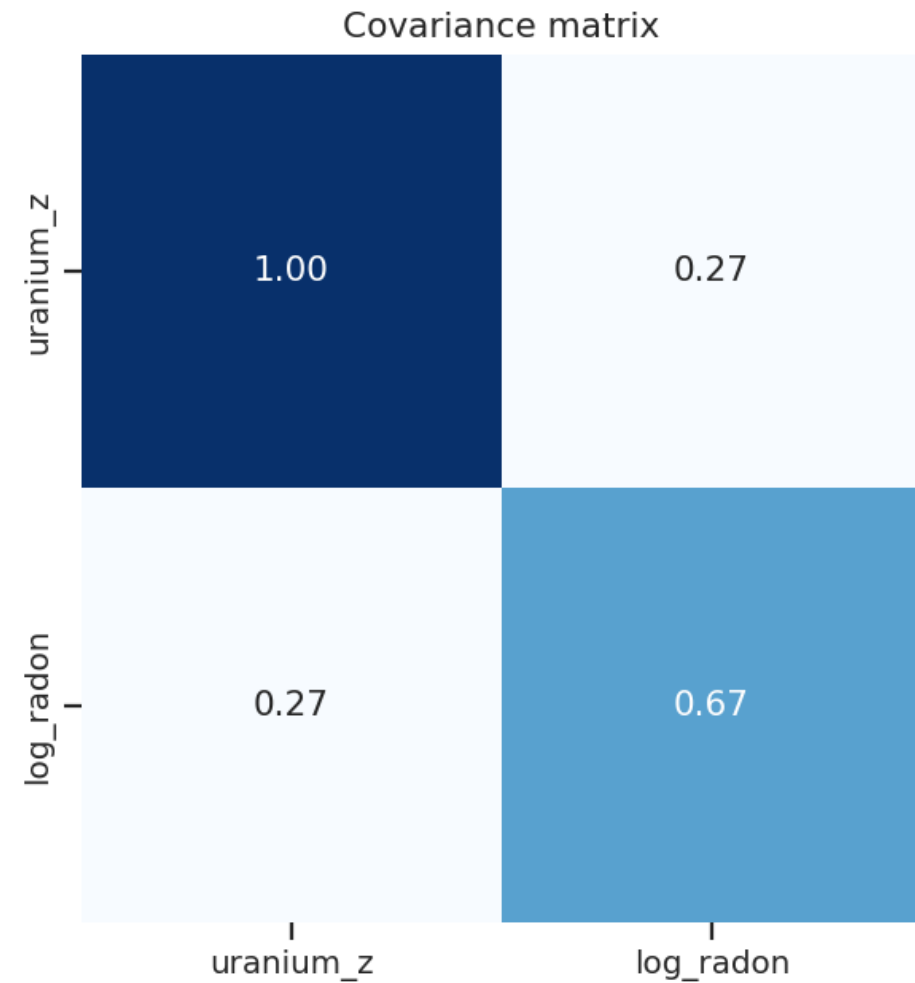
$$\Sigma = \begin{bmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{bmatrix}$$

- Diagonal entries: variances.
- Off-diagonal entries: covariance.
- The matrix is symmetric.
- Covariance depends on units.

Empirical matrices

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix}$$

- Diagonal entries: variances.
- Off-diagonal entries: covariance.
- The matrix is symmetric.
- Covariance depends on units.



Bayesian correlation model

$$[x_i, y_i]^T \sim MVNormal(\mu, \Sigma)$$

$$\mu = [\mu_x, \mu_y]^T$$

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix}$$

- ρ appears inside the likelihood through the covariance matrix.
- A univariate Normal likelihood cannot estimate correlation between two outcomes.
- The model estimates μ , σ , and ρ jointly.

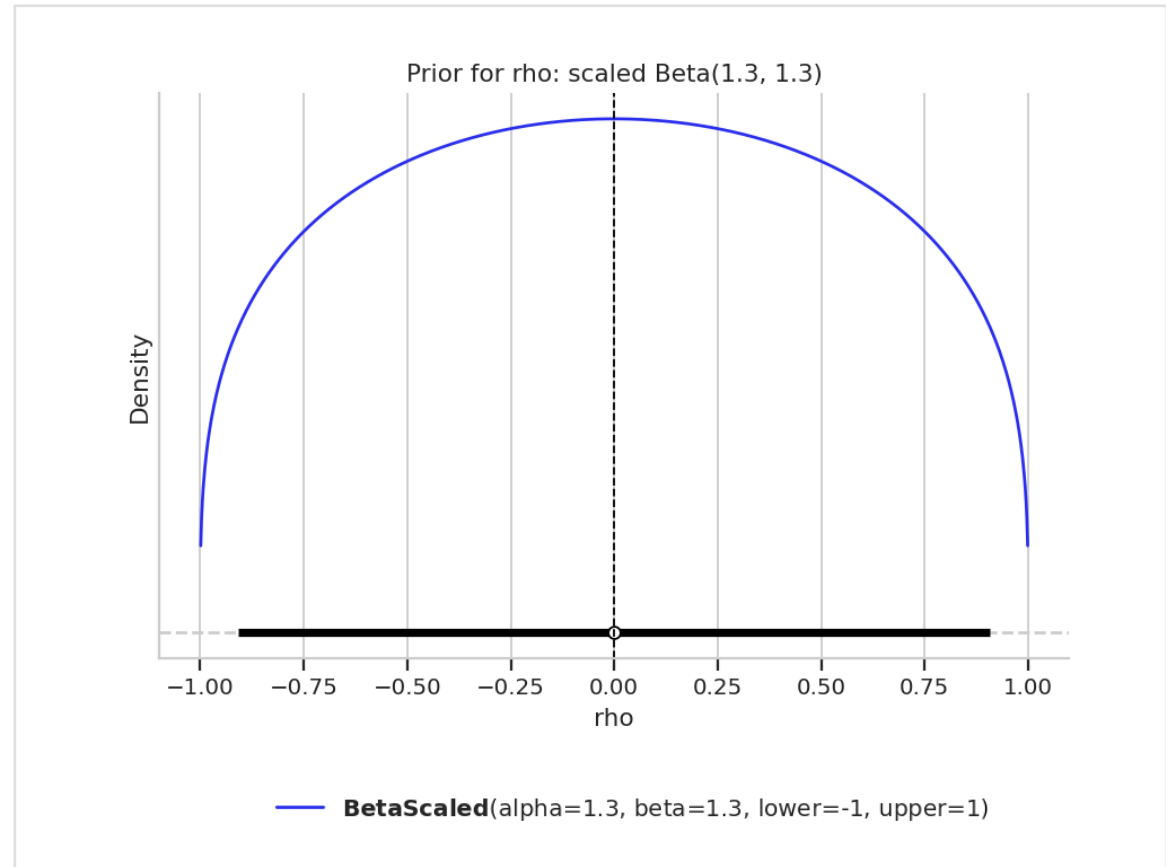
Prior for ρ

$$\theta_\rho \sim \text{Beta}(1.3, 1.3)$$

$$\rho = 2\theta_\rho - 1$$

- A Beta distribution lives on $[0, 1]$.
- The transformation maps it to $[-1, 1]$.
- This gives a simple bounded prior for correlation.

```
theta_rho = pm.Beta("theta_rho", alpha=1.3, beta=1.3)
rho = pm.Deterministic("rho", 2 * theta_rho - 1)
rho_squared = pm.Deterministic("rho_squared", rho ** 2)
```



Prior density for ρ .

Bayesian correlation model

$$[x_i, y_i]^T \sim MVNormal(\mu, \Sigma)$$

```
xy = correlation_data.to_numpy()

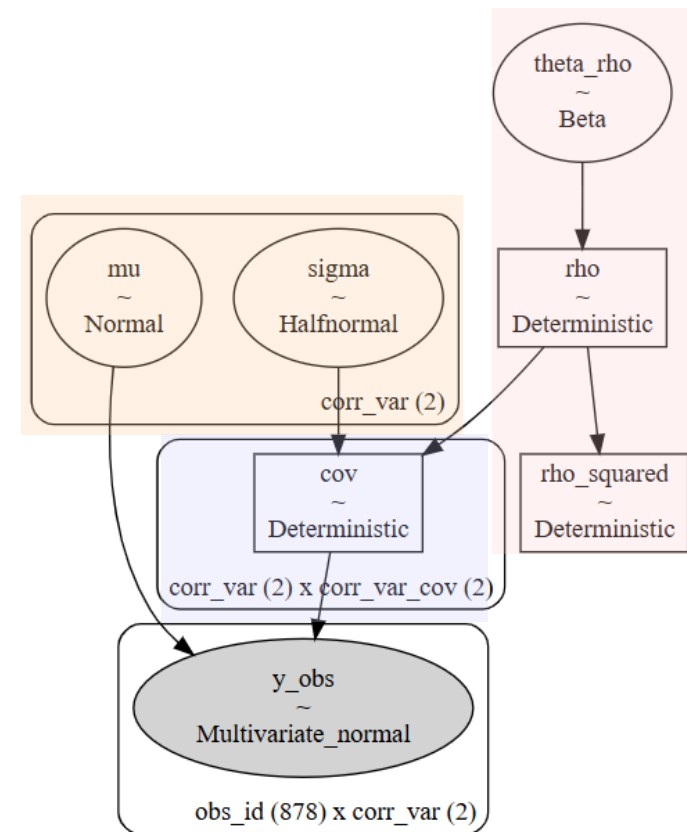
coords_corr = {"obs_id": np.arange(len(xy)),
               "corr_var": ["uranium_z", "log_radon_z"],
               "corr_var_cov": ["uranium_z", "log_radon_z"],}

with pm.Model(coords=coords_corr) as correlation_model:
    mu = pm.Normal("mu", mu=0, sigma=1, dims="corr_var")
    sigma = pm.HalfNormal("sigma", sigma=1, dims="corr_var")

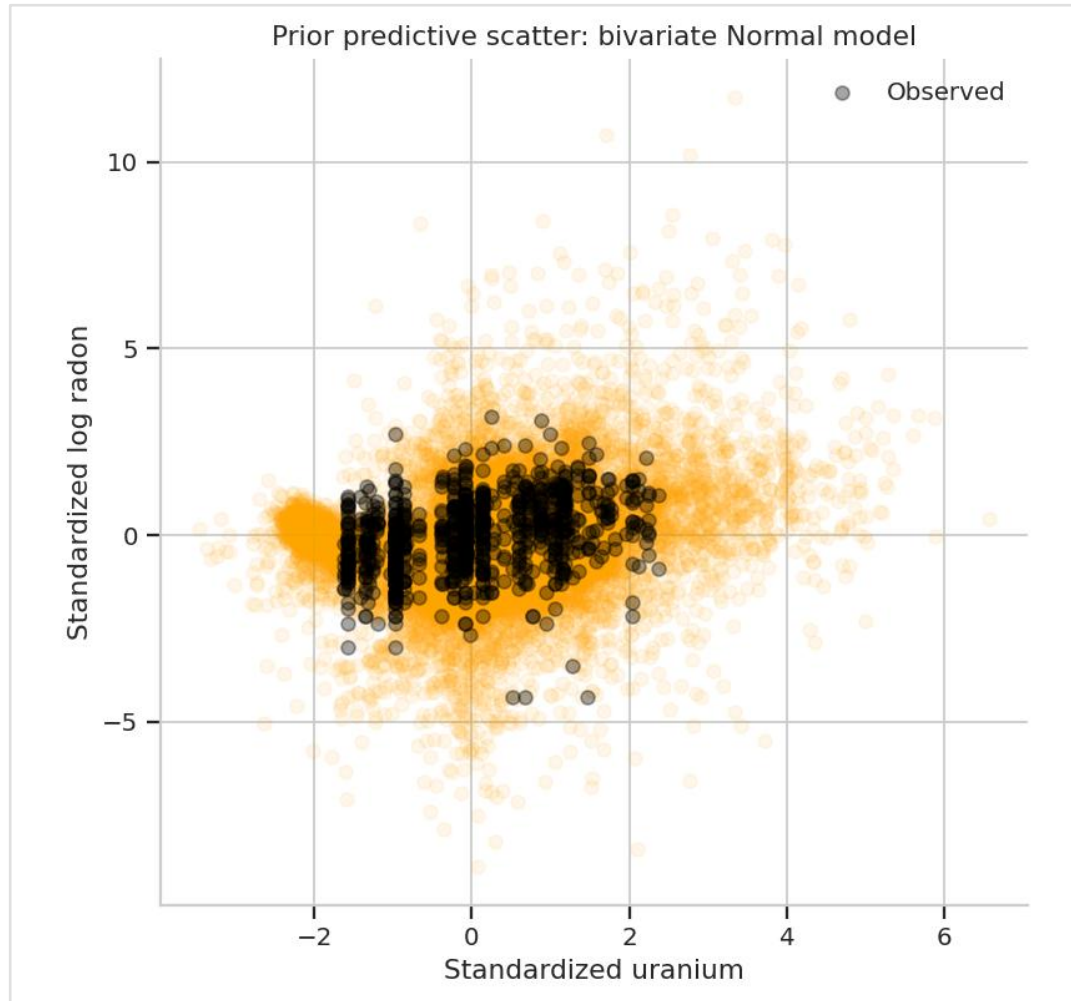
    theta_rho = pm.Beta("theta_rho", alpha=1.3, beta=1.3)
    rho = pm.Deterministic("rho", 2 * theta_rho - 1)
    rho_squared = pm.Deterministic("rho_squared", rho ** 2)

    cov = pm.Deterministic(
        "cov",
        pt.stack([
            pt.stack([sigma[0] ** 2, rho * sigma[0] * sigma[1]]),
            pt.stack([rho * sigma[0] * sigma[1], sigma[1] ** 2]),
        ]),
        dims=("corr_var", "corr_var_cov"))

    y_obs = pm.MvNormal(
        "y_obs",
        mu=mu,
        cov=cov,
        observed=xy,
        dims=("obs_id", "corr_var"))
```



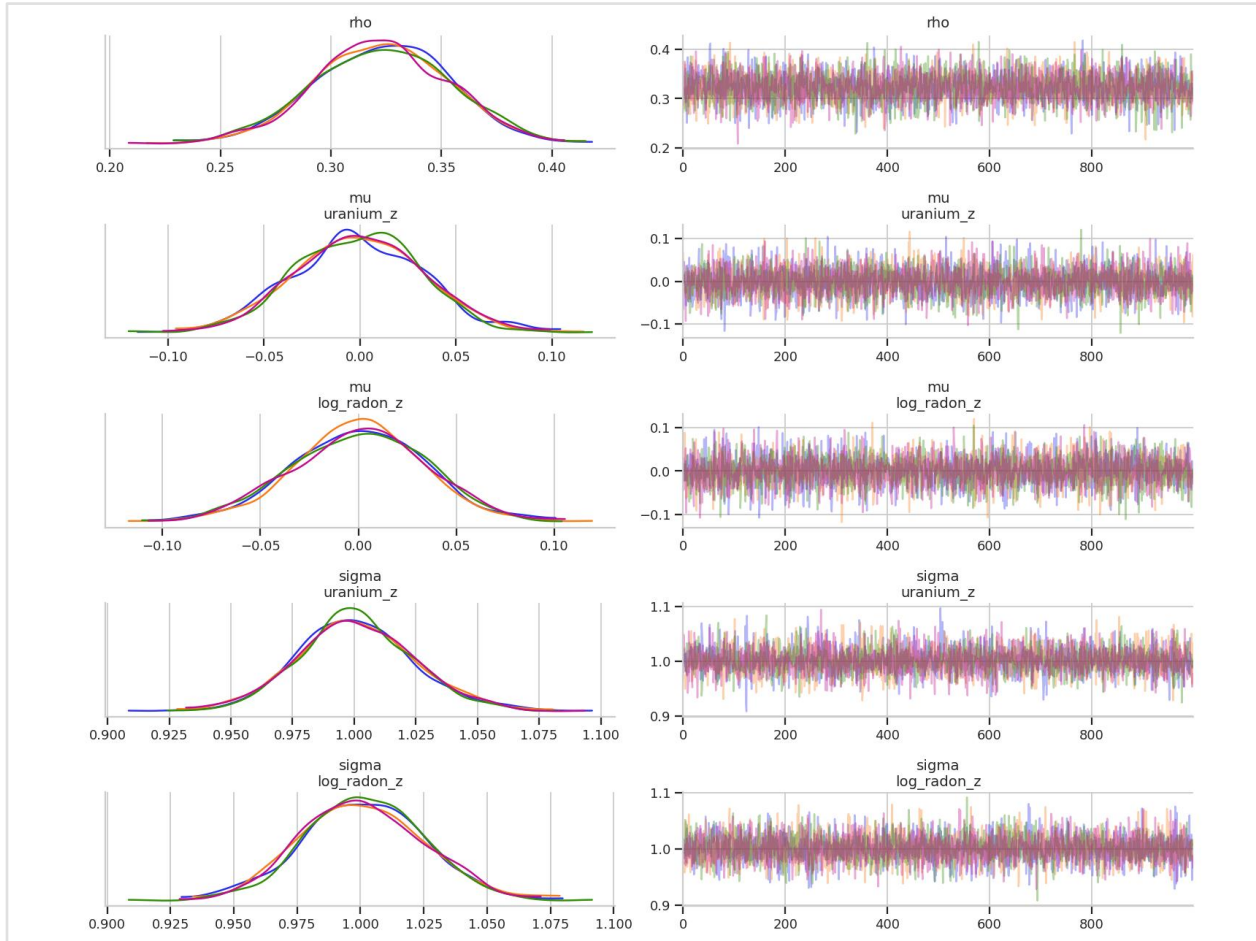
Bayesian correlation: prior predictive



Prior predictive scatter for the bivariate Normal model.

- Each orange point cloud is a dataset drawn from the prior predictive distribution.
- Observed data are overlaid in black.
- The prior predictive check asks whether the joint model is plausible before fitting.

Bayesian correlation: diagnostics

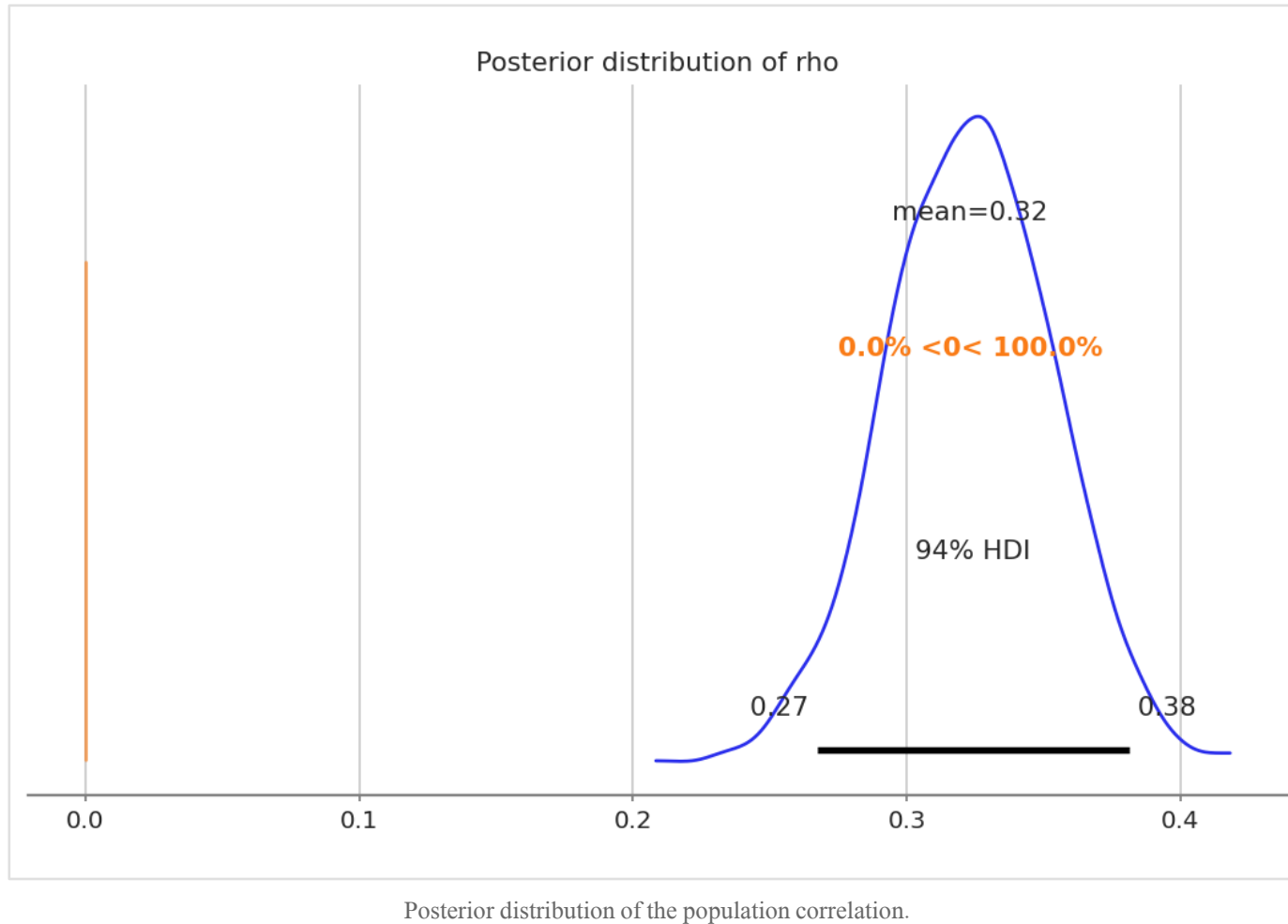


Trace plots for the bivariate Normal model.

Parameter	ESS bulk	R-hat
ρ	4045	1.00
ρ^2	4045	1.00
$\mu[\text{uranium}^z]$	4375	1.00
$\mu[\text{log_radon}^z]$	4253	1.00

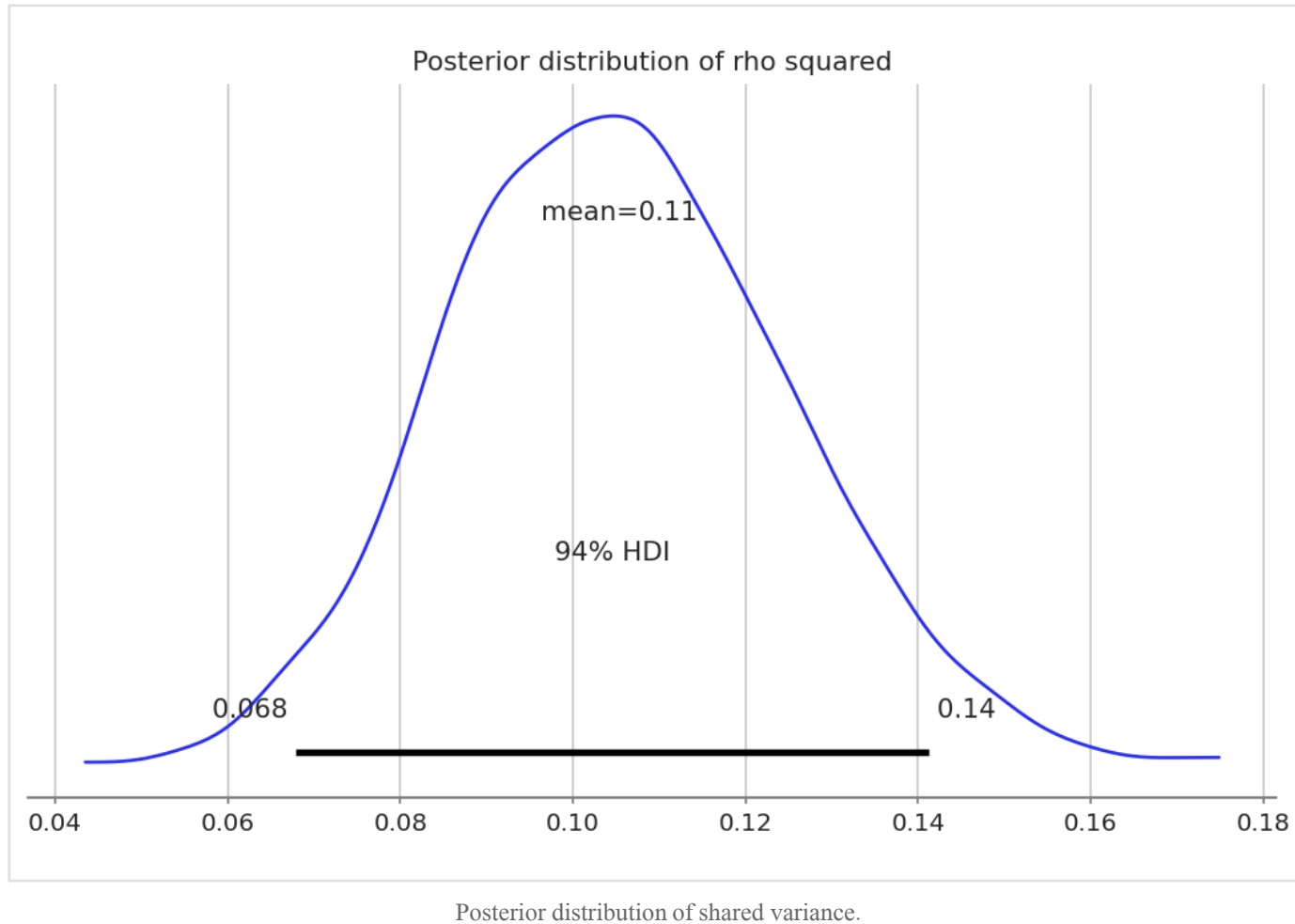
Diagnostics support interpreting the posterior for ρ and ρ^2 .

Posterior distribution of ρ



- Posterior mean $\rho \approx 0.323$
- 94% HDI: [0.267, 0.381].
- The posterior is clearly positive in this model.
- This describes association, not causation.

Posterior distribution of ρ^2



- Posterior mean $\rho^2 \approx 0.105$.
- 94% HDI: [0.068, 0.141].
- Interpretation: roughly 10% shared variance in the linear association sense.
- Shared variance describes association not a causal effect.

Pearson correlation versus Bayesian correlation

Quantity	Value
Pearson r	0.324
Posterior mean ρ	0.323
ρ 94% HDI	[0.267, 0.381]
Pearson r^2	0.105
Posterior mean ρ^2	0.105
ρ^2 94% HDI	[0.068, 0.141]

- Pearson r is a sample statistic.
- The Bayesian model estimates a posterior distribution for ρ .
- The posterior directly expresses uncertainty about plausible population correlations.
- The p-value asks a different question: compatibility with $H_0: \rho = 0$.

Final conceptual summary

Grouped dependence:

houses in the same county may be more similar than houses in different counties.
Hierarchical regression represents this using county-level intercepts.

Symmetric association:

two continuous measurements can vary together without a predictor/outcome direction.

ρ is the population correlation coefficient, and it measures the direction and strength of the linear relationship between two variables.

ρ^2 , the squared correlation, measures the proportion of shared linear variance between the two variables.

The bivariate Normal model represents this through the covariance matrix and ρ .
The modeling choice should follow the scientific question and data structure.